

Simultaneous multiclass retinal lesion segmentation using fully automated RILBP-YNet in diabetic retinopathy

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ABSTRACT

This paper presents a novel method for segmenting the multiple lesions like Microaneurysms, Haemorrhages, Hard and Soft exudates that occurs on the surface of the retina due to diabetes. If these lesions are left untreated in their early stage, they may cause irrevocable vision loss to diabetic patients. However, the manual assessment of these vision threatening diseases is prone to errors and time-consuming process. As a result, the proposed model, RILBP-YNet (Residual Inception Local Binary Pattern - YNetwork) is implemented as a dual encoder and single decoder architecture in which one encoder extracts the features from the fundus image. Whereas the parallel encoder extracts the textural features from Local Binary Patterns (LBPs) of the corresponding fundus image simultaneously. The extracted features of the two encoders are concatenated at the end of every encoder level and fed as input to the next encoder. Finally, the concatenated features of the last encoder are transposed by performing the up-sampling operation and fed as input to the first decoder. Furthermore, at the end of every decoder, the features of the decoder and concatenated features of the respective encoder are fused and provided as input to the next decoder. As a result, at the end of the last decoder, segmented masks of each lesion are generated. The novelty of the proposed work lies in adding a parallel encoder to traditional encoder-decoder architecture to induce textural features to improve the segmentation results. The proposed model is trained and tested on the Indian Diabetic Retinopathy Image Dataset (IDRiD), and e-optha datasets and achieved competitive results compared to the existing methods in terms of AUC score. The proposed method is also tested on datasets like DRIVE, STARE, CHASE-DB1, and one locally real-time dataset.

1. Introduction

Diabetic Retinopathy (DR) is an eye condition that is developed in people with diabetes mellitus. It is caused by damage to the blood vessels present on the retina of the eye and can lead to severe conditions like blurred vision, and vision loss, and can be fatal in extreme conditions. DR can be broadly categorized into two categories: Non-Proliferate Diabetic Retinopathy (NPDR) to Proliferate Diabetic Retinopathy (PDR). The NPDR is the early stage of DR in which as an initial sign, red lesions called Microaneurysms (MA) are formed on the retinal surface. Further, DR and its other complications [12] causes other lesions like Haemorrhages (HE), Hard Exudates (EX), and Soft Exudates (SE) as shown in Fig. 1. If these initial symptoms were not treated earlier, they advance to the severe stage termed as Proliferate Diabetic Retinopathy (PDR). In PDR, new fragile blood vessels are formed and termed

Neovascularization. These vessels get ruptured easily and cause irreversible damage to the retina of the eye.

To circumvent this lacuna, the early detection of these lesions is very essential in the diagnosis of DR and its other complications like Diabetic Macular Edema (DME), etc. Several clinical and scientific attempts have been made to improve the process of screening DR [34]. Different imaging techniques are used to screen DR patients. Fluorescein Angiography (FA) is an invasive technique where fluorescein is injected into the eyes to examine the vascular map of the fundus but there are side effects like severe headache, hypotension, vomiting, etc. B-scan ultrasonography is used to generate the image of a retina by capturing reflected sound waves transmitted at high frequency, but it is used for diagnosing cataracts, retinal detachments, etc. Color fundus photography is a non-invasive technique for capturing the retinal surface of a patient and is being widely used recently as they are easy to interpret. OCT can

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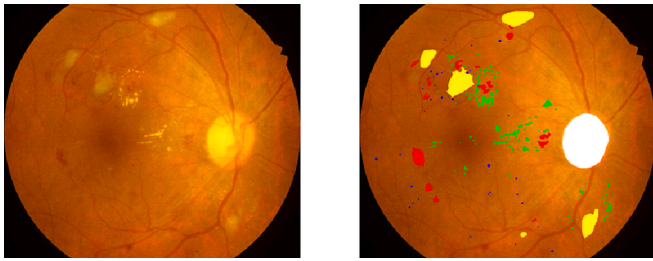


Fig. 1. A sample image from IDRiD dataset with original fundus image (left) and its ground truth annotation (right). The colors blue, red, green, yellow, and white represent MA, HE, EX, SE, and OD respectively.

estimate the thickness of different retinal layers and can help to detect multiple retinal disorders, but it is very expensive. Among various techniques available, fundus photography is considered a better alternative for screening DR as it is cost-efficient and available even in some low and middle-income countries. In the analysis of the type of diagnosis, the screening of retinal disease plays a vital role. Easy and accessible diagnosis is necessary for the early detection of this global epidemic.

According to a study conducted by International Diabetes Federation (IDF) in 2019 [5], involving 7000 adults with diabetes and healthcare professionals from 41 countries around the world, it is reported that 79 % of the respondents who had their vision impaired due to DR made their everyday activities difficult. Also, a total population of 463 million adults (20–79 years) worldwide was living with diabetes, and 93 million among them are estimated to have DR. One-third of the global diabetic population is expected to develop some form of retinopathy in their lifetime. People suffering from DR and Diabetic Macular Edema (DME) contribute to 1 % of blindness and visual impairment worldwide. The global prevalence of DR and DME for the period of 2015 to 2019 was 27 % for any type of DR, 25.2 % for NPDR, 1.4 % for PDR, and 4.6 % for DME. The diabetic population around the world is expected to rise to 700 million by 2045 [5]. While DR is considered a global problem, 75 % of the diabetic population lives in low and middle-income countries with inadequate healthcare facilities. The National Diabetes and Diabetic Retinopathy Survey of India [6] in 2019 reported that the prevalence of any form of DR in the diabetic population aged up to 50 years was found to be 16.9 %, whereas the prevalence of blindness was 2.1 % and vision impairment was 13.7 %. As of March 2020, diabetes is prevalent in 8.9 % population, i.e., 77,005,600 adults in India, and it is expected to rise to 153 million by 2045.

The manual examination of retinal images is a long-time process for expert doctors/ ophthalmologists. To differentiate the lesions, experts need DR-specific knowledge, but as IDF [5] reported 21 % of ophthalmologists i.e., 1 in 5 ophthalmologists, have not received any specific training on diagnosis and treatment of DR. Apart from this, many low and middle-income countries suffer from the unavailability of equipment pertaining to the screening of DR. Computer-aided diagnosis assists experts, where an intelligent system like a deep learning model mimics the experts for rapid segmentation of different retinal lesions. All these alarming statistics motivated this work in developing a robust computer-aided diagnosing system to detect all these early signs of DR & its complications and prevent further loss of visual damage to the eyes. Some of the challenges faced during the process of segmenting the lesions are listed below:

1. **Multi-Class Imbalance:** A common challenge in the segmentation of lesions is observed due to intra and inter-class imbalances in the training dataset. The intra-class imbalance occurs due to the low prevalence of lesion pixels compared to background non-lesion pixels of fundus images. Furthermore, the pixel count of one class of lesion varies with the pixel count of other class lesions. i.e., for example, in the IDRiD dataset, the average pixel count of MA on

resized fundus image is 5, whereas the average pixel counts of HE, EX, and SE of resized fundus image are 20, 8, and 34 respectively as shown in Table 1. These differences in the pixel count of each lesion cause intra-imbalance in the dataset. Besides, these lesions are also not present in every image of the dataset. i.e., in IDRiD [7] dataset, lesions such as MAs, and EXs are present in every training image (54 out of 54), but HEs are present in 53 training images, and SEs are present only in 26 out of 54 training images. This results in the interclass imbalance of the dataset and eventually affects the training performance of the proposed model.

2. **Size and Shape of the Lesions:** The size and shape of each type of lesion vary with each image. Due to this, it is a cumbersome task to generate recursive patterns to detect each lesion individually from the testing dataset.
3. **Color of the Lesions:** The interclass similarity index between the lesions is very high. i.e., the visual appearance of the lesion classes (MA & HE) and (EX & SE) are very similar and hard to distinguish with human eyes. As a result, it is a challenging task to segment between various classes with the manual grading dataset.

To overcome the above-mentioned challenges, in this paper, we propose a model that performs semantic segmentation on the fundus image and generates multiple segmentation masks for each class of lesions along with the optic disc of the fundus image. The proposed model RILBP-YNet is implemented in the dual encoder and single decoder style for the task of multi-class segmentation of lesions by incorporating the local textural information using Local Binary Patterns (LBP). The main motivation of the model is to detect all four lesions along with the optic disc by incorporating textural patterns via parallel encoder to detect the DR and its complications in their early stage. The major contributions of the proposed model are as follows:

1. The proposed RILBP-YNet is developed with two encoders to extract the feature maps and precisely detect the lesions from fundus images. To mitigate the lacunas of conventional U-Net in extracting the most pertinent features. The proposed RILBP-YNet employed two parallel encoders in which one extracts the conventional convolutional features from input images and other extracts the feature maps from textural patterns of corresponding input image enabling the network to focus on lesion regions more attentively. The textural patterns highlight the lesion pixels and enhances the intensity values such that there will be high variance in terms of intensity between lesion and non-lesion pixels. In the proposed model, to estimate the textural features, Local Binary Patterns (LBPs) [8] are utilized as they are invariant to different image augmentation techniques like scaling, rotation, etc, and empowers the model to detect the irregular and complex patterns of different objects.
2. The encoders of the network are driven with Residual and Inception (RI) modules, where the inception module is used to extract multi-scale features by convoluting the fundus image with kernels of varying sizes (1×1), (3×3), (5×5). Simultaneously, the residual modules are used to transfer feature maps between different stages of convolutions. Further, at the end of each encoder, the textural feature maps obtained from the local binary patterns of the fundus images are convoluted and concatenated with the feature maps obtained from the RI block and fed as input to the consequent encoder to enrich the features with textural information of input images. This empowers the network to detect lesions more effectively.
3. In addition to this, a novel weighted combination of the loss function is employed to update the weights of the network enabling to detection of the lesion pixels more efficiently.
4. The raw fundus images obtained from different public and one local real-time datasets are enhanced by using different pre-processing techniques like contrast enhancement, gamma correction, etc to improve the quality of the input images. Further, to reduce the class imbalance, different data augmentation techniques like cropping,

Table 1

Pixel count details of each lesion on fundus images of IDRiD dataset.

Type of Lesion	Original Image			Resized fundus Image		
	Minimum	Maximum	Average	Minimum	Maximum	Average
MA	5	76	23	1	17	5
HE	4	3252	101	1	655	20
EX	4	1645	36	1	379	8
SE	4	700	166	1	105	34

rotation, random cropping, etc are applied to increase the strength of the training dataset.

- To evaluate the efficiency of the proposed model, it is trained and tested on IDRiD [7] and e-optha datasets. It is observed that the proposed RILBP-YNet shows competitive results compared to the table-toppers of the “IDRiD: Diabetic Retinopathy - Segmentation and Grading Challenge” [9] by achieving AUC scores as 0.6914 (proposed) vs. 0.6804, and 0.7267 (proposed) vs. 0.7113 for HEs, and SEs respectively. The proposed method outperformed existing methods in segmenting HEs, EXs, and SEs on the e-optha dataset and scored the highest mAUC score. Furthermore, to estimate the performance of the proposed RILBP-YNet, it is tested on different public and local datasets with the same weights obtained from the IDRiD dataset.

The remaining part of the manuscript is organized as follows: Section 2 describes the literature review of the proposed work along with the lacunas encountered in each method. Then, explained the importance of the proposed model in overcoming the lacunas encountered by earlier methods along with a description of each module of the proposed model in Section 3. Section 4 includes the experimental details that illustrate the training and testing performance of the proposed RILBPY-Net by performing different comparative and ablation studies. Further Section 5, the discussion section highlights the advantages of the proposed model in segmenting the lesions. Finally, Section 6 concludes with a summary of the novel proposed model along with its future improvements and applications.

2. Related work

Earlier, research has been done on detecting DR in traditional methods by extracting hand-crafted features from fundus images, performing feature selection, and classifying whether a person has DR and then grading. Apart from this, retinal blood vessel analysis can be done for screening PDR, and extensive research was done on blood vessel segmentation [10–12]. Later research has progressed by applying supervised [13–20] and unsupervised learning [21–24] techniques to detect DR. Features like the area of the lesion, the intensity of pixels, mean, variance, etc are calculated. Different feature selection techniques are applied to optimize the extracted features. Different machine learnings classifiers like K Nearest Neighbours (KNN) [15,16], Support Vector Machine (SVM) [17–19], Naïve Bayes classifier [20], etc. are trained using the optimized features to classify retinal images of DR into respective grades. Apart from this, different unsupervised algorithms like fuzzy C-means clustering [22], and K-means clustering [23,24] are used for lesion segmentation. The method proposed by Roy et al. [17] performs similar work of extracting features like area, distance from the optic disc, intensities of pixels, etc, then selecting the top k features based on their contribution to classification. This method achieved 100 % sensitivity and 53.16 % specificity in classifying DR into different grades. But it is a repetitive process to check for different combinations of input features and finding the correct combination of top k features that contribute more to the classification. Lachure et al. [25] proposed a method in which different features like morphological, Gray Level Co-occurrence Matrix (GLCM), and structural features are extracted from the pre-processed image. Two different classifiers, SVM and KNN are

trained and achieved 100 % specificity, 90 % sensitivity when tested on a private dataset. Although, it is evident that hand-crafted feature extraction and selection have achieved better sensitivity they are difficult to implement and are time-consuming processes.

With advancements in Artificial Intelligence, many researchers have adopted deep learning techniques and created deep learning models that extract relevant features and update themselves in the process of training. The development of Deep Convolutional Neural Networks (DCNNs) [26] has replaced the process of hand-crafted feature extraction and feature selection. These models have significantly sped up the process of classification/segmentation by improving their performance when compared with the traditional methods. Recently different deep learning models have been developed to segment medical images like UNet [27], UNet++ [28], 3D-UNet [29], CAB-UNet [30], S-UNet [31], and Capsule Networks [32]. Even retinal vessel segmentation is automated by employing neural networks as implemented by Wu et al. [33]. Jiang et al. [34] proposed a multi-path recurrent UNet that takes the shape of a 3D network with skip connections present between encoders and decoders. The 3D UNet segments the optic disc, optic cup, and retinal blood vessels by achieving 99 % specificity and 93 % sensitivity. Tan et al. [35] developed a 10-layer convolutional neural network that performs the segmentation of MAs, HEs, and exudates (both EX and SE) in fundus images. They achieved a sensitivity of 87 % for exudates and 71 % for dark lesions, which include HEs and MAs on the CLEOPATRA dataset. An improved UNet, RIC-UNet [36], is developed where the power of Residual and Inception Modules are combined with UNet to deliver higher performance for the segmentation of histology images. This method has achieved a Dice Coefficient of 0.8008 and an F1-score of 0.8278 and outperformed the traditional UNet. Many semantic segmentation networks have been developed like SegNet [37], which performs the segmentation of scenes using a modified decoder by calculating pooling indices during the max-pooling process in the encoder. This method has achieved an average class accuracy and an average global accuracy of 71.2 % and 90.4 %, respectively which is better than the state-of-the-art models like FCNN [38], DeepLab-LargeFOV [39], and DeconvNet [40]. All the above methods either perform classification tasks that categorize the fundus images to DR and Non-DR cases or detect only a particular lesion-related DR. There is no specific semantic segmentation network developed earlier to the best of our knowledge that can detect every lesion related to complications of DR along with OD segmentation.

The IDRiD Diabetic Retinopathy – Segmentation and Grading Challenge [9] accepted models from different researchers who performed the task of segmenting the fundus images. The top performer, VRT, used a modified UNet that has the same number of feature maps after upsampling for segmenting the lesions, and they reported a mean Area Under Curve (mAUC) score of 0.6469. The second top performer, iFLYTEK-MIG, also used an UNet in a novel three-stage segmentation, coarse segmentation, cascade classifier for false-positive reduction, and fine segmentation of lesions. Then the team PATech developed a modified UNet combined with DenseNets and dilated blocks. The drawback observed in all the competitors is that the same architecture is used with varying depths and input shapes for segmenting specific types of lesions. While testing, it is difficult to load separate models to segment each lesion for a single image. In view of developing a single deep learning model that performs multi-class segmentation, inspired by the model

developed by Guo et al. [2], we have developed a model that segments multiple lesions at a time. Cheng Wan et al. [41] developed EAD-Net that extracts the pertinent features using a dual attention module. Caixia et al. [42] developed a model by combining U-Net++ with the residual module that enables the network to segment the lesions more accurately by regaining the relevant low-level features. Guo et al. [43] implemented a cascade attentive RefineNet (CARNet) by combining U-Nets. Whereas the features of every decoder of one U-Net are concatenated with the respective decoder of other U-Net to attain accurate segment masks. Marlin et al. [44] designed a lightweight U²-Net for the segmentation of multiple lesions that reduces the memory burden of U-Net. To reduce the uncertainties of the deep learning model, Azat et al. [45] modified the Bayesian framework by considering the parameters of CNN. However, this method has not completely succeeded in overcoming the challenges of deep learning models. The proposed model performs semantic segmentation on the fundus image to segment different lesions present in the fundus image at a single iteration. Our novelty lies in combining both the self-learning features and handcrafted features uniquely by incorporating two parallel encoders. Local Binary Patterns are good texture descriptors and have proven to effectively segment DR lesions proposed by Y. He et al. [21].

3. Methodology

Our model RILBP-YNet is trained on the IDRiD, and e-optha datasets by considering the fundus image and corresponding LBPs of the enhanced green channel as inputs. The process of performing semantic segmentation using RILBP-YNet is divided into three stages: 1) pre-processing of the fundus image, 2) calculating the local binary pattern of the green channel of the image, and 3) RILBP-YNet input and generating masks.

3.1. Pre-processing

The pre-processing of the RGB fundus image is shown in Fig. 2. The original fundus image obtained from IDRiD [7] dataset has a black background with unwanted information. It is cropped out by performing contour analysis, then resized to a fixed dimension (512, 576, 3) to generalize the input shape of the deep learning model. Let us consider the input fundus image as $IN(x, y)$, which is composed of three channels, Red $IN_r(x, y)$, Green $IN_g(x, y)$, and Blue $IN_b(x, y)$. It is observed that the green channel of the fundus image has better contrast compared to the others. As a result, the green channel of the input image is extracted and further enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE) with a grid size of (8, 8) and a clip limit of 1.5 as per equation (2). In CLAHE, the entire image is split into non-overlapped grids of size (8, 8), and applies histogram equalization to each tile for better contrast. The clip limit of CLAHE determines the maximum contrast enhancement that can be applied on each grid of the input image. In addition to this, to control the overall brightness of the

enhanced image and mitigate the over-exposure conditions, gamma correction (GAMMA) is performed as per equations (3) and (4) respectively. The enhanced green channel is used to calculate the local binary patterns.

$$IN(x, y) = [IN_r(x, y), IN_g(x, y), IN_b(x, y)] \quad (1)$$

$$\{IN_g(x, y) \text{ has better contrast}\}$$

$$IN_{g-clahe} = CLAHE(IN_g, clipLimit = 1.5, gridSize = (8, 8)) \quad (2)$$

$$IN_{g-clahe-gamma} = GAMMA(IN_{g-clahe}, \gamma = 1.5) \quad (3)$$

where,

$$GAMMA(L, \gamma) = \left\{ \left(\frac{L}{L_{max}} \right)^\gamma \right\} * L_{max} \quad (4)$$

3.2. Local binary Patterns:

Local Binary Patterns [8] (LBP) is a feature descriptor that can capture local texture information enabling the model to detect objects of different shapes more effectively. These LBPs are invariant to image augmentation techniques like scaling, rotation, etc, and empowers the model to detect the irregular and complex patterns of different objects. In the proposed work, the LBPs are calculated from the pre-processed retinal image with four different radii to generate four different local binary pattern representations of the image to feed into the LBP encoder. LBPs are calculated using radii of 1, 3, 5, and 7 to extract patterns of different scales from the fundus image. LBPs are computed by comparing each pixel with the neighbourhood pixels. A pixel 'N' and a radius 'R' is selected from the image, after that, a set of 'P' pixels from the neighborhood of the N at a distance R are used to calculate the LBP of that pixel. Every pixel i from the set of P pixels is compared with the value of N and a boolean value is assigned accordingly, i.e., either 0 or 1. If the i th pixel is greater than the selected pixel, it is assigned with 1, else 0 is assigned. The binary sequence is converted to a decimal number and is assigned to the pixel N as per equation (5).

$$LBP_{P,R} = \sum_{i=0}^{P-1} S(N - P_i) * 2^i \quad (5)$$

where

$$S(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

To achieve rotation invariance, the uniform local binary patterns are calculated with the utmost two 0-1 or 1-0 transitions present in the binary sequence as per equation (6).

$$LBP_{P,R} = \begin{cases} \sum_{i=0}^{P-1} S(N - P_i) * 2^i, & T(LBP_{P,R}) \leq 2 \\ P + 1, & \text{otherwise} \end{cases} \quad (6)$$

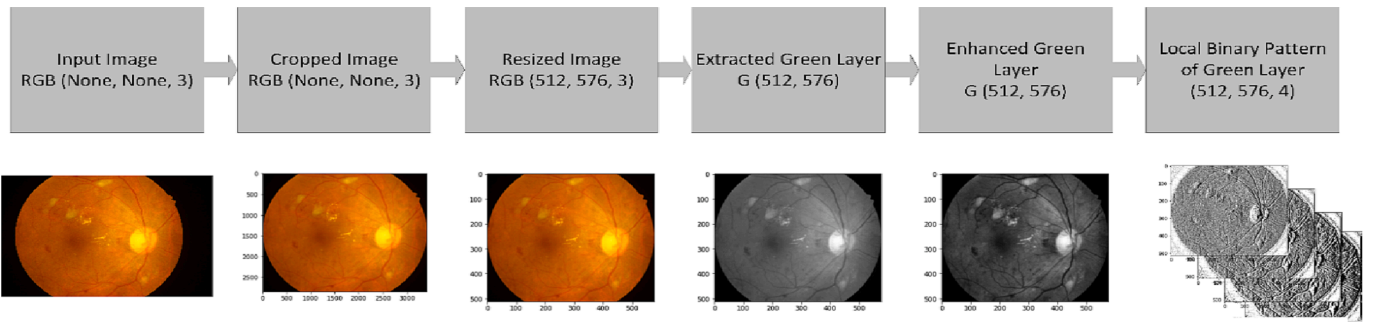


Fig. 2. The pre-processing flow of the fundus image before feeding to the network. A series of operations are performed on the RGB fundus image and LBPs are calculated on the enhanced green channel of the fundus image.

where $T(LBP_{P,R})$ denote the 0–1 or 1–0 transitions in the binary sequence.

The uniform local binary patterns are calculated with four different values of $R = 1, 3, 5 \& 7$, and $P = 1$.

3.3. RILBP-YNet

The proposed model RILBP-YNet takes two inputs because of its dual encoder architecture, feeding one with resized fundus image and the corresponding parallel encoder with four LBPs of the enhanced green channel at once, as shown in Fig. 3. The RILBP-YNet variant has a depth of 4.

3.3.1. Encoder implementation

The encoders of the network are driven with Residual-Inception (RI) blocks as shown in Fig. 4, and the implementation of the RI block is shown in Fig. 5. All the convolution operations performed in the network use a stride of (1, 1) and are initialized using the ReLU activation function. The Residual module is used to maintain the flow of information between different levels of the convolution layers. The Inception module performs convolutions with kernels of different sizes, (1*1), (3*3), (5*5), and a MaxPooling layer of kernel size (3*3) to extract multi-scale features to overcome the challenge associated with lesions of different shapes and sizes. The extracted feature maps are concatenated, followed by convolution with a kernel of size (1, 1). Both the Inception and Residual feature maps are added and feed-forward to the deeper levels preceded by a ReLU activation. The first level of the encoder generates 32 feature maps, and the feature maps get doubled as we go deeper into the network as 64, 128, and 256. At every level of the network, the feature maps obtained from the LBP encoder are concatenated with those of the fundus image encoder after MaxPooling with a stride of (2, 2). The resultant feature maps are fed to the deeper levels of the encoder. Finally, a dropout layer is applied with a probability of 0.5. The feature maps obtained from convoluting the texture patterns of the green channel can be used to guide the encoder to extract relevant features.

3.3.2. Decoder implementation

In the decoder, the convolution operations use a stride of (1, 1) and are initialized using ReLU activation. The 256 feature maps at the last level of the encoder are up-sampled, followed by Batch Normalisation. The resultant feature maps are concatenated with the feature maps from

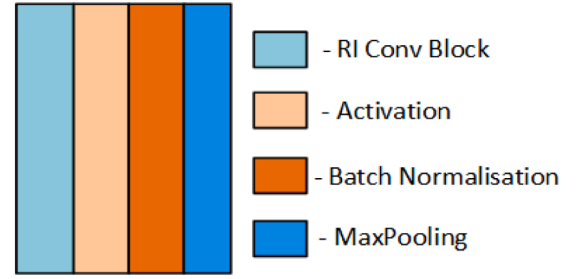


Fig. 4. Encoder block of RI-LBP-YNet.

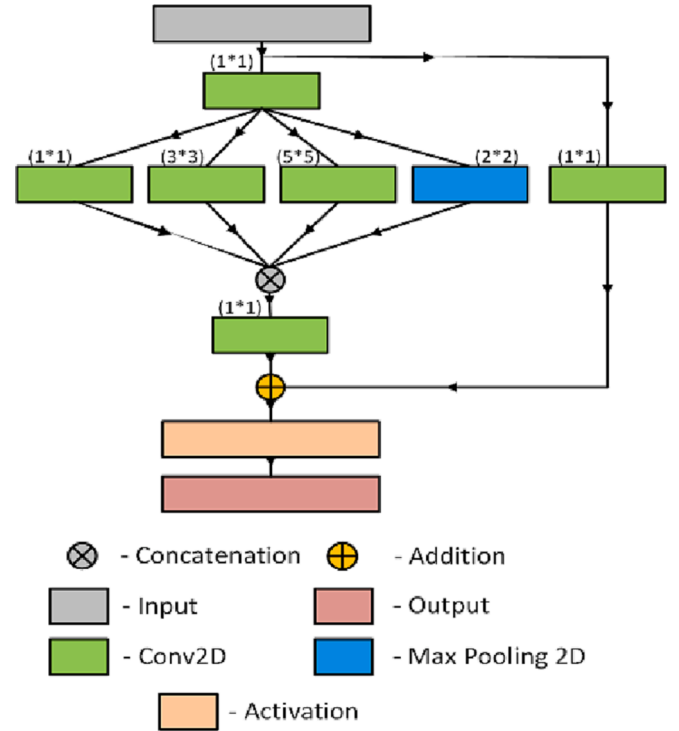


Fig. 5. Residual - Inception Block of Encoder.

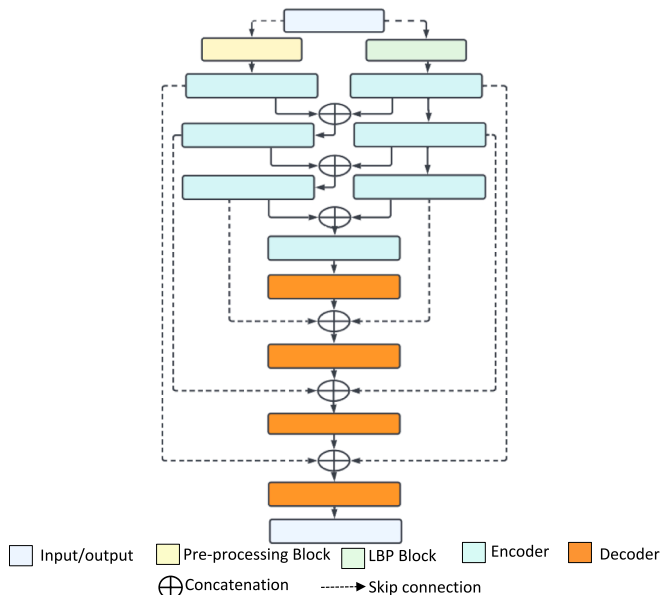


Fig. 3. The entire framework of RILBP-YNet architecture with depth 4.

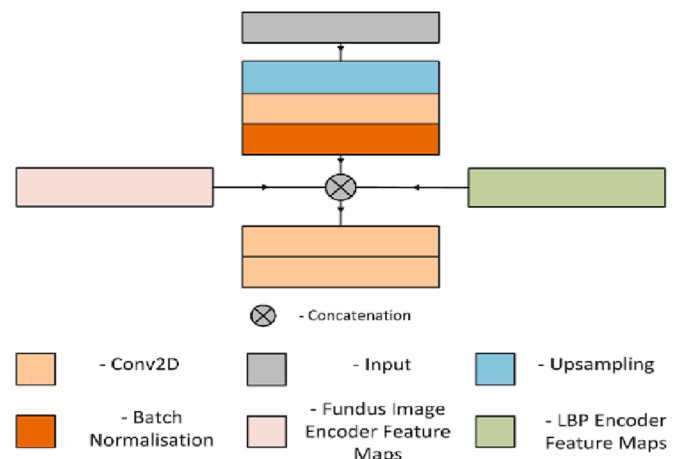


Fig. 6. Decoder block of RILBP-YNet.

both the encoders of the corresponding stage along the depth axis. A couple of convolution operations are performed to generate feature maps of depth 128, as shown in Fig. 6. Then a series of up-sampling and convolution operations are performed on the resultant feature maps. Finally, the feature maps of the two encoders and feature maps of the decoder are concatenated at each stage to increase the dimensions of the feature maps to match with input image size. Ultimately, the final segmented masks are obtained by generating feature maps of depth 5, i.e., each feature map represents one class of retinal lesions.

3.3.3. Loss function

The RILBP-YNet model is trained using a loss function L_{com} i.e., a weighted combination of two-loss functions. The loss functions, Jaccard Loss (JL) and Categorical Cross-Entropy (CCE) combined in the ratio 2:1. The JL is a measure of loss due to the non-overlapping regions of ground truths and segmentation masks, whereas the CCE loss is a measure of loss due to misclassified pixels of each class as per equations (7–9). Finally, the loss of each class is added together to update the weights of the proposed RILBP-YNet.

$$JL(y, p) = 1 - \frac{\sum_{n=0}^{N-1} \sum_{i=0}^{H*W-1} y_{ni} * p_{ni}}{\sum_{n=0}^{N-1} \sum_{i=0}^{H*W-1} y_{ni} + p_{ni} - y_{ni} * p_{ni}} \quad (7)$$

$$CCE(y, p) = -\frac{1}{N} \sum_{n=0}^{N-1} \sum_{j=0}^{H*W-1} \{y_{ni} * \log p_{ni} + (1 - y_{ni}) * \log(1 - p_{ni})\} \quad (8)$$

The combined loss of the proposed model is defined as L_{com} and is estimated by using equation (9).

$$L_{com} = \alpha * JL(y, p) + \beta * CCE(y, p) \quad (9)$$

Where y is the ground truth, and p is the predicted value, N is the number of classes, H is the height of the ground truth or predicted mask, W is the width of the ground truth or predicted mask. α and β are the scaling parameters in the range of $[0, 1]$ and are considered in the ratio 2:1.

4. EXPERIMENTAL RESULTS

4.1. Dataset

The proposed model RILBP-YNet is trained and evaluated on two publicly available datasets, i.e., IDRiD and e-optha. It is also tested on three publicly available datasets, i.e., DRIVE, STARE, CHASEDB-1, and one locally collected real-time dataset. The IDRiD[7] dataset is publicly available as a part of a challenge hosted by the International Symposium of Biomedical Imaging (ISBI) conference in 2018. This challenge aims to develop a robust computer-aided diagnosis tool for the screening of DR that reduces the cost burden and can ease the process of screening in a mass country population. The segmentation dataset consists of 81 fundus images and the pixel-level annotated labels for MA, HE, EX, and SE. The images have a resolution of 4288*2848 (W*H) and 50° Field of View (FOV). The dataset is provided as training and testing sets, with 54 images for training and 27 images for testing. The e-optha [46] dataset consists of images having a resolution of 2544*1696 and 40° FOV with pixel-level annotations for two different lesions, i.e., EX for 34 images and MA for 68 images. DRIVE [47] dataset consists of 40 fundus images with a resolution of 584*565 and 45° FOV of which 33 are normal, and 7 have mild early DR present in them. STARE [48] dataset comprises 31 normal and 50 pathologically affected images. The size of each image in this dataset is 700*605 with 35° FOV. CHASEDB-1[49] has 28 fundus images with a resolution of 1280*960. In addition to this a local dataset containing 18 real time fundus images having resolution 3504*2336 are collected from Sankar Foundation Eye Hospital, Visakhapatnam and disease-based camps conducted in rural areas of Andhra Pradesh, India for evaluating the proposed model on real data. The result of local dataset is validated by the team of expert ophthalmologists from department of vitreous retina, of our collaborating hospital, Sankar

Foundation Eye Hospital, Visakhapatnam.

4.2. Implementation of the proposed RILBP-YNet:

To enhance the input images, different pre-processing techniques like cropping, histogram equalization, and gamma correction are applied to the extracted green channel of the fundus image, as shown in Fig. 7. The different datasets used for training and testing of the proposed network have different characteristics i.e., they consist of images having different resolutions. Also, datasets like IDRiD and e-optha have unwanted black background present in the images whereas the other datasets DRIVE, STARE, CHASEDB-1, and the local dataset doesn't have a black background. The pre-processing steps are modified according to the dataset while giving input to the model either by applying or removing the cropping operation. In addition to this, histogram matching is applied to the images of DRIVE, STARE, CHASEDB-1, and local datasets using a reference image from the IDRiD dataset. Apart from this, other pre-processing techniques like CLAHE and gamma correction with a gamma value of 1.5 are applied to all the images irrespective of the dataset to enhance and control the intensity values for better quality images. On experimenting with different combinations of grid size and clip limit for applying CLAHE, it is observed that a grid size of 8×8 and clip limit of 1.5 enhanced the fundus images more efficiently than other values by achieving a Peak Signal to Noise Ratio (PSNR) value of 28 dB. Further, the gamma correction mitigates the issues of over brightness that occurred in some images after applying CLAHE on resized input images. To further enlarge the training dataset, five types of data augmentation techniques are applied to the training images and respective ground truth masks, i.e., top to bottom flip, left-to-right flip, 90 deg rotation, 270 deg rotation, and 80 % random cropping of the fundus to create 324 training images out of the original 54 images from IDRiD dataset.

The augmented images are then resized to a fixed dimension (512, 576, 3) and fed to the network in batches of 4 images. For each batch of fundus images, corresponding local binary patterns are calculated as shown in Fig. 8 and given as input to the parallel encoder simultaneously. The edges of lesions and blood vessels present in the fundus image are differentiated more precisely using LBPs. As a result, the exudates are more apparent and guide the fundus image encoder in segmenting lesions. The network is optimized using Adam optimizer with a learning rate of 0.0001. If no decrement in loss values for seven consecutive iterations is observed, then the learning rate is reduced by a factor of 0.1. The model is monitored on its metrics, like loss and dice coefficient at each epoch, and trained for 50 epochs. The weights of the model with the least loss are loaded into the network and it is again trained for further 50 epochs, this process is repeated to attain better segmentation results. The model is trained for a maximum of 100 epochs with early stopping based on monitoring the Dice Coefficient, and a dropout of 0.5 probability is applied to prevent the network from overfitting. The proposed method, RILBP-YNet having 4.8 M trainable parameters is implemented using TensorFlow 2.0 framework and Keras library on publicly available cloud GPU services, Google Collaboratory, and Kaggle Notebooks. With the inclusion of LBP's as parallel input, the performance of the model raised by a factor of 5 % in segmenting MAs & HEs and 10 % in segmenting EXs & SEs in terms of Dice Coefficient.

4.3. Evaluation metrics

To evaluate our model, we have used similar evaluation metrics used in the IDRiD challenge, Area Under the Precision-Recall (AUPR) curve. The curve was obtained by thresholding at 33 equally separated intervals in the range of 0 to 1, i.e., $[0, 0.03125, 0.0625, \dots, 1]$ in probabilities [9].

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

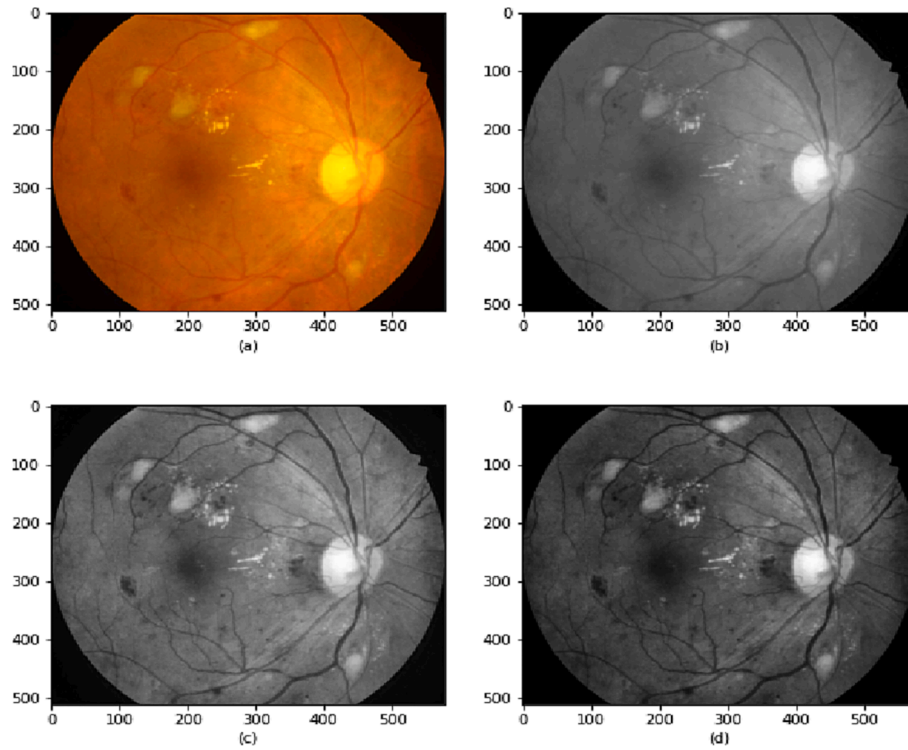


Fig. 7. The process of pre-processing a sample image of IDRiD dataset, (a) fundus image, (b) extracted green channel, (c) CLAHE applied green channel and (d) Gamma Corrected resultant image.

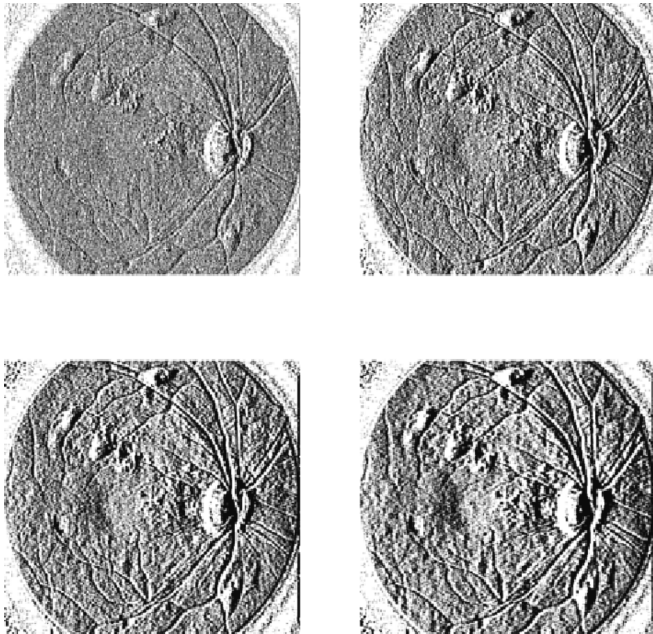


Fig. 8. Local Binary Patterns calculated on the enhanced green channel of sample fundus image with different radii 1, 3, 5 & 7 from left to right and top to bottom.

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

Where *TP* indicates True Positives, *FP* indicates False Positives, and *FN* indicates False Negatives.

We have also evaluated our model using the Dice Coefficient, which is a measure of the area of overlap between two masks.

$$DiceCoefficient(y, y') = \frac{2 * |y \cap y'|}{|y| + |y'|} \quad (14)$$

4.4. Segmentation Results:

The proposed RILBP-YNet generates 5 probabilistic maps for each lesion along with the optic disc. A common threshold index of 0.5 is applied to the obtained probabilistic maps to deliver 5 binary masks for all classes i.e., MA, HE, EX, SE, and OD respectively. The precision-recall curves are plotted for every type of lesion present in the classes, as shown in Fig. 9. The Area Under the Precision-Recall curve is calculated for every type of lesion in the IDRiD dataset, 0.3804 for MA, 0.6914 for HE, 0.7858 for EX, 0.7267 for SE. The model is trained and monitored by evaluating its performance metrics like Dice Coefficient, Loss, Sensitivity, and Specificity. The metrics and loss values monitored while training the model are presented in Fig. 10. At the end of the training, an average dice coefficient of 0.63 is observed, and the loss is 0.59 with a sensitivity of 0.87 and specificity of 0.99. To illustrate the quantitative performance of the proposed model, the proposed model is tested on the IDRiD dataset for each lesion individually as per [60] and the results are depicted in Table 2.

The box plots for each type of lesion considered from the IDRiD dataset against their sensitivities are plotted as shown in Fig. 11. The average sensitivity of the optic disc is exceptionally high with a value of 0.95, followed by Hard Exudates with 0.74, Soft Exudates with 0.48, Haemorrhages with 0.42, and Micro-Aneurysms with 0.22. The average sensitivity among all the types of lesions is calculated to be 0.73. Some outliers were observed while segmenting MAs because of their smaller size, i.e., the area contributed by them at a range of one to ten pixels. Similarly, SEs are spread over a large region. As a result of this, a large class imbalance is present in the dataset. The dice coefficients for each lesion present in the testing set are calculated, and the results are shown in Fig. 12. The proposed model scored a dice coefficient of 0.434 for MA, 0.525 for HE, 0.751 for EX, 0.7860 for SE, and 0.963 for OD, and the

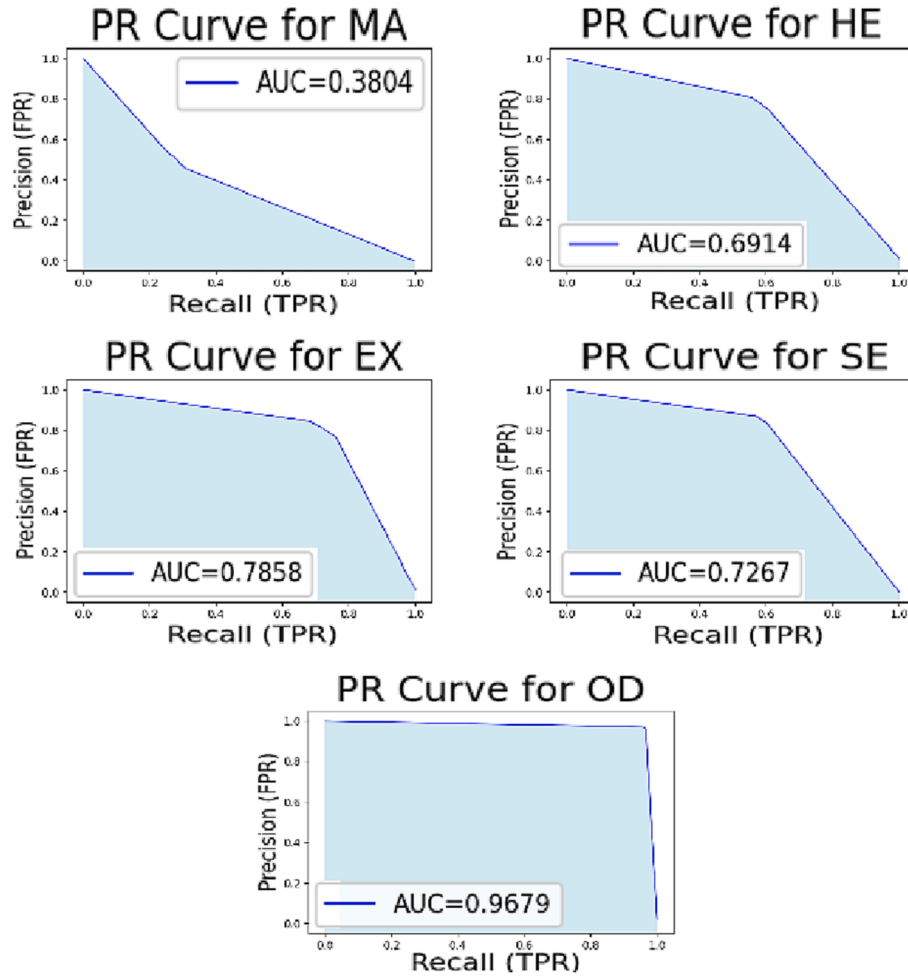


Fig. 9. Precision - Recall curves of lesions, Micro - Aneurysms, Haemorrhages, Hard Exudates, Soft Exudates, and Optic Disc with Area Under PR Curve calculated on the IDRiD dataset.

results of the IDRiD dataset are visualized in Fig. 12.

The model is evaluated on the IDRiD dataset, and semantic segmentation results for sample images are plotted in Fig. 13. The proposed model can segment different lesions along with optic disc segmentation. The activation maps depict the formation of lesions after each up-sampling operation in the decoder of the network. The progression of lesion segmentation in the decoder blocks at different levels of the RILBP-YNet is visualized in Fig. 14.

4.5. Comparison of RILBP-YNet with IDRiD challenge teams

The results of our model, RILBP-YNet on the IDRiD dataset compared with the top teams of the IDRiD challenge [9] in Table 3. The proposed model has scored an AUC of 0.7267 for SE and stood at 1st position, 0.6914 for HE and stood at 1st position, and scored an mAUC score of 0.6586 and stood at 1st position. Further, as per [9], these top teams of the IDRiD challenge methods implemented separate networks for each lesion whereas the proposed model alone efficiently segmented all the lesions along with the optic disc by achieving an mAUC of 65.8 %.

4.6. Comparison of segmentation results of RILBP-YNet with baseline models on various datasets

The proposed model is compared with the baseline models, L-seg [2], FCN [50], HED [51], CASENet [52], and DeepLab v3+ [53] on the IDRiD dataset, and the results are shown in Table 4. Our model stood first in segmenting hemorrhages, soft exudates, and hard exudates. Also,

the proposed model outperformed the previous networks in terms of mAUC. Furthermore, Fig. 15 visually illustrates the supremacy of the proposed model over other models. In addition to this, the proposed model is compared with the state-of-the-art algorithms using the IDRiD dataset, and the respective results are depicted in Table 5. To evaluate the efficacy and vulnerability of our model, RILBP-YNet is trained and tested on the e-optha dataset [46], which has two sub-datasets for single lesion segmentation, EX, and MA. The proposed RILBP-YNet model scored an AUC score of 0.5147 for EX, and 0.2092 for MA. An mAUC score of 0.3399, as shown in Table 6. Further, due to the unavailability of standard pixel level annotated images, the proposed model has been qualitatively tested on datasets like DRIVE, CHASE, STARE, and local datasets (real-time captured images) using the weights of the IDRiD dataset. To generalize the input characteristics of the testing image, initially, the testing images are subjected to histogram matching with the sample image of the IDRiD dataset and fed as input into the network. The respective results are depicted in Fig. 16. From Fig. 16., it is observed that the proposed model is more accurately segmenting OD than lesions due to intensity variations between datasets. These testing results are validated manually by the expert doctors and estimated that the proposed model attained an average dice coefficient of 95 % for OD, 70 % for SE, 75 % for EX, 50 % for HE, and 35 % for ME respectively. The proposed RILBP-YNet suffers from false positives in segmenting the lesions, especially for MA and HE. The limitation of the RILBP-YNet towards segmenting a specific class of lesions will be mitigated by training the model with the various fundus of different datasets.

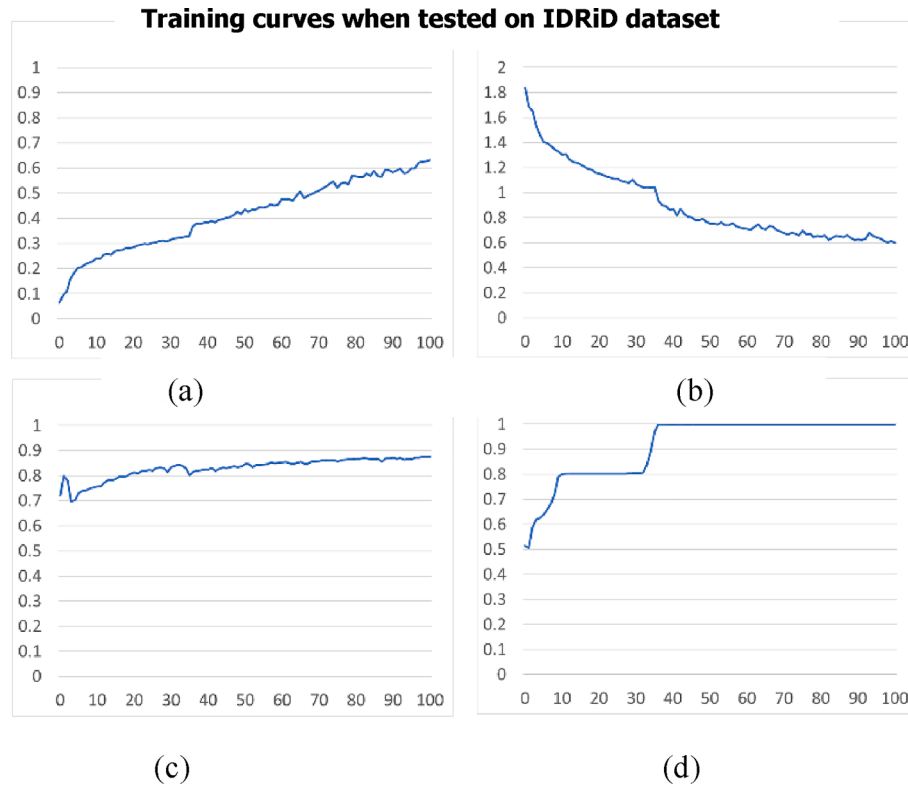


Fig. 10. Training progress of RILBP-YNet in terms of (a) Dice Coefficient, (b) Loss, (c) Sensitivity, and (d) Specificity vs Epochs.

Table 2

Quantitative parameters of proposed RILBP-YNet for each lesion on IDRiD dataset [9] with a common threshold index of 0.5.

Regions	Se	Sp	Acc	AUPR	F1
MA	28.02	99.96	99.87	38.04	36.02
HE	57.95	99.69	99.00	69.14	66.54
EX	72.15	99.77	99.42	78.58	76.32
SE	58.55	99.96	99.85	72.67	49.88
OD	96.2	96.03	99.80	96.2	96.1

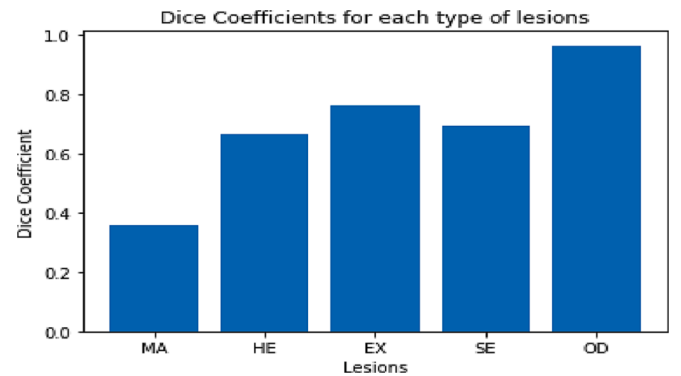


Fig. 12. Dice Coefficients for different types of lesions and optic disc for testing set of IDRiD dataset.

Sensitivity values of each class when tested on IDRiD dataset

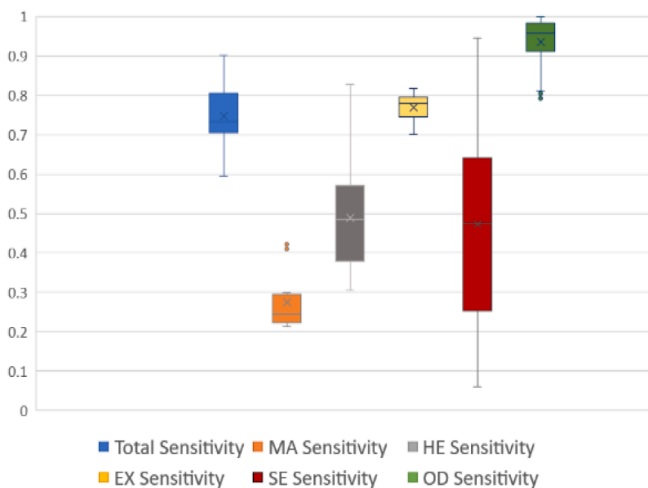


Fig. 11. Box Plots of Sensitivities of each class when tested on IDRiD dataset.

4.7. Abaltion studies

The novel combination of residual and inception modules in the network along with the parallel encoder of local binary patterns enabled the proposed RILBP-YNet to segment the lesions more efficiently. To illustrate the impact of each component, ablation studies are performed, and results are tabulated in Table 7 in terms of the dice coefficient, parameter size, and training time. It is observed that the inception modules raised the performance of the model in segmenting hard and soft exudates. Further, the residual modules combine the low-level features with high-level features to segment smaller lesions like micro-aneurysms more accurately. From Fig. 17 it is visually clear that the local binary patterns from the parallel encoder enhanced the lesion textures and enabled the network to detect the lesions more accurately. Overall, the combination of residual and inception modules with the proposed model increased the dice coefficient by 5 %. Whereas the LBPs along with other pre-processing techniques raised the performance of

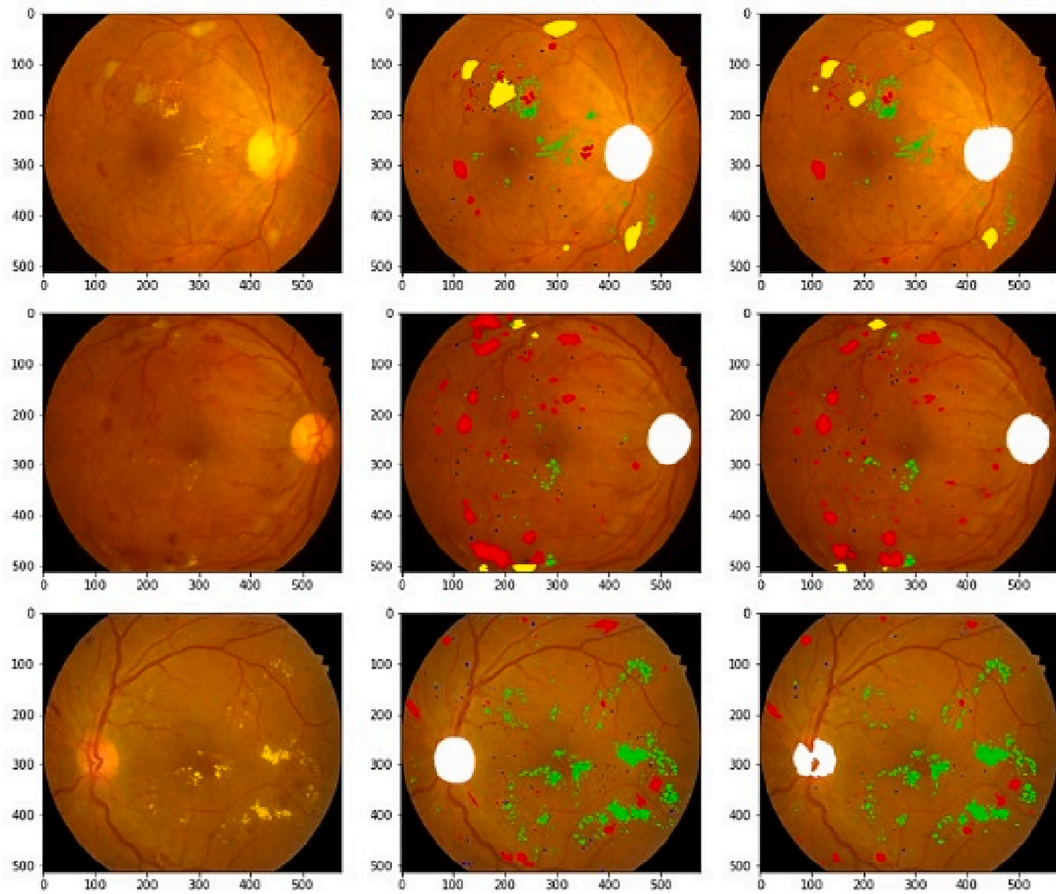


Fig. 13. Images from IDRiD dataset, with ground truth annotations and predicted masks (from left to right).

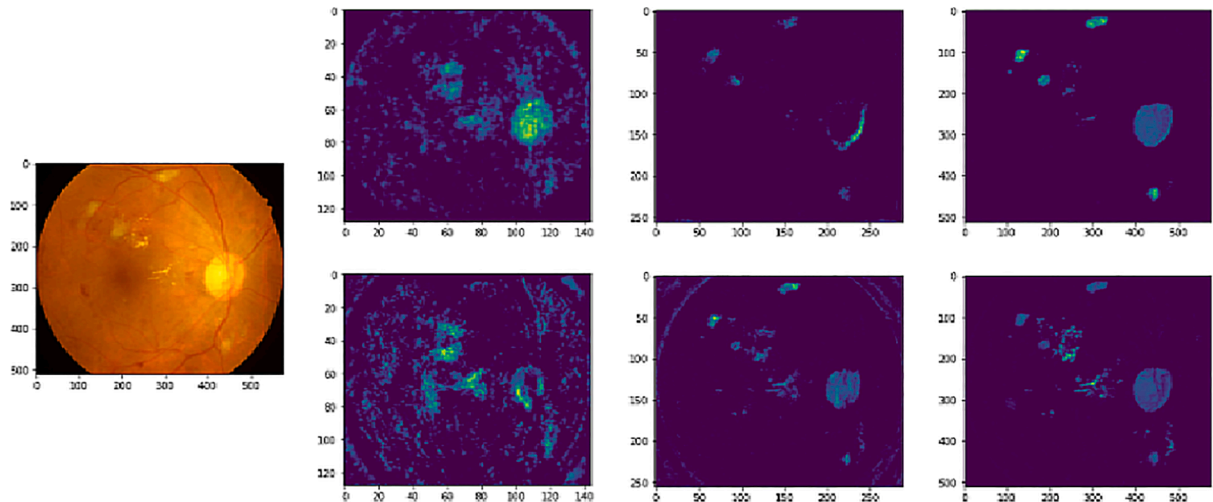


Fig. 14. The feature maps of convolution layers in decoder from higher stages to lower stages (left to right), before and after adding feature maps from LBP encoder (top to bottom).

RILBP-Ynet by 5 to 8 % more than Y-Net [58], and LBP-Net [59]. Further studies were conducted to understand how the changes in loss function impact the performance of the RILBP-YNet and results are depicted in Fig. 18. Table 8 represents the performance improvement in the monitored training metrics with respect to each loss function used with the proposed architecture. A significant improvement in the dice coefficient of MA can be seen from 0.2810 to 0.434 with the use of the weighted loss function of categorical cross entropy and Jaccard loss rather than using

categorical cross entropy alone. In addition to that, it can be observed from Table 8 that there has been an improvement in dice coefficients while segmenting all the lesions from the fundus image of the IDRiD dataset. The use of the weighted loss function raised the dice coefficient of each lesion by at least 10 % more than jaccard loss values.

Table 3

Comparison of AUC scores of the proposed method with IDRiD Challenge Teams as per [9]. (* Bold values indicate the highest values.).

Network (Team)	MA	HE	EX	SE	mAUC
VRT	0.4951	0.6804	0.7127	0.6995	0.6469
PATech	0.474	0.6490	0.8850	–	–
iFLYTEK-MIG	0.5017	0.5588	0.8741	0.6588	0.6484
SOONER	0.4003	0.5395	0.7390	0.5369	0.5539
SAIHST	–	–	0.8582	–	–
lzyunc fusion	–	–	0.8202	0.6259	–
SDNU	0.4111	0.4572	0.5018	0.5374	0.4769
CIL	0.3920	0.4886	0.7554	0.5024	0.5346
MedLabs	0.3397	0.3705	0.7863	0.2637	0.4401
AIMIA	0.3792	0.3283	0.7662	0.2733	0.4367
L-Seg	0.4627	0.6374	0.7945	0.7113	0.6515
RILBP-YNet	0.3804	0.6914	0.7858	0.7267	0.6586

Table 4

Comparison of AUC scores of the proposed method with Baseline Models as per [2]. (* Bold values indicate the highest values.).

Network	MA	HE	EX	SE	mAUC
L-seg [2]	0.4627	0.6374	0.7945	0.7113	0.6515
HED [51]	0.4271	0.5084	0.7723	0.6673	0.5938
FCRN [50]	0.3383	0.4200	0.5472	0.5154	0.4552
CASENet [52]	0.3992	0.4462	0.7564	0.3275	0.4823
DeepLab v3+ [53]	0.1514	0.4772	0.7118	0.5912	0.4829
RILBP-YNet	0.3804	0.6914	0.7958	0.7267	0.6586

5. Discussion

Negligence of DR in its early stage progresses to severe eye-related complications. The primary screening of lesions like MA, HE, EX, and SE prevents visual impairment. This paper presents several contributions like combining textural features with self-learning features and a novel method of adding a parallel encoder to the traditional encoder-decoder architecture for multi-class lesion segmentation on fundus images. In the proposed pre-processing steps, the green channel of the fundus images is used to generate LBPs and provide input to the parallel encoder.

The raw images in the IDRiD dataset have a resolution of 4288*2848. Due to the limited availability of the GPUs and further to maintain homogeneity in the implementation of the network, the input raw images are cropped and resized to a resolution of 576*512. As a result, the lesion pixels are also rescaled to a lower number of pixels. There are lesions of different sizes and shapes. Of all those lesions, MAs are spread

over a very small area in pixels. The quantitative analysis of various lesions before and after resizing images is performed on the IDRiD dataset, and the results are shown in Table 1. Due to the small and irregular structure of multi-class lesions, it is challenging to segment them simultaneously. However, the incorporation of Inception and Residual modules in the encoder block of the proposed network has significantly improved its performance in segmenting the lesions. The quantitative comparison between the implementations, i.e., the serial structure of convolutional layers and the parallel structure of convolution layers based on the inception module, is shown in Table 9. The capability of segmenting every lesion has been improved due to the usage of the kernels of different sizes (1*1), (3*3), and (5*5) in the parallel structure to extract multi-scale features rather than the serial structure of convolution layers with single kernel size (1*1). As a result,

Table 5

Comparison with existing lesion segmentation networks on IDRiD dataset [9]. (* Bold values indicate the highest values.).

Method	Lesion	Se	Sp	Acc	AUPR	F1
Sarhan et al. [54]	MA	28.07	–	–	41.96	38.4877
Guo et al. [2]		–	–	–	46.27	–
Yan et al. [55]		–	–	–	52.5	–
Xue et al. [56]		76.4	99.8	99.7	–	–
Proposed		28.02	99.96	99.87	38.04	36.02
Guo et al. [2]	HE	–	–	–	67.34	–
Yan et al. [55]		–	–	–	70.3	–
Proposed		57.95	99.69	99.00	69.14	66.54
Guo et al. [2]	EX	–	–	–	79.45	–
Yan et al. [55]		–	–	–	88.9	–
Xue et al. [56]		77.9	99.6	99.2	–	–
Proposed		72.15	99.77	99.42	78.58	76.32
Guo et al. [2]	SE	–	–	–	71.13	–
Yan et al. [55]		–	–	–	67.9	–
Proposed		58.55	99.96	99.85	72.67	49.88

Table 6

Comparison with baseline models on the e-optha dataset in terms of AUC values. (* Bold values indicate the highest values.).

Network	EX	MA	mAUC
HED [51]	0.1345	0.0639	0.0992
FCRN [50]	0.1675	0.0269	0.0972
CASENet[52]	0.1715	0.1565	0.1640
DeepLab v3+ [57]	0.5512	0.0045	0.2778
L-Seg [2]	0.4171	0.1687	0.2929
RILBP-YNet	0.5147	0.2092	0.3399

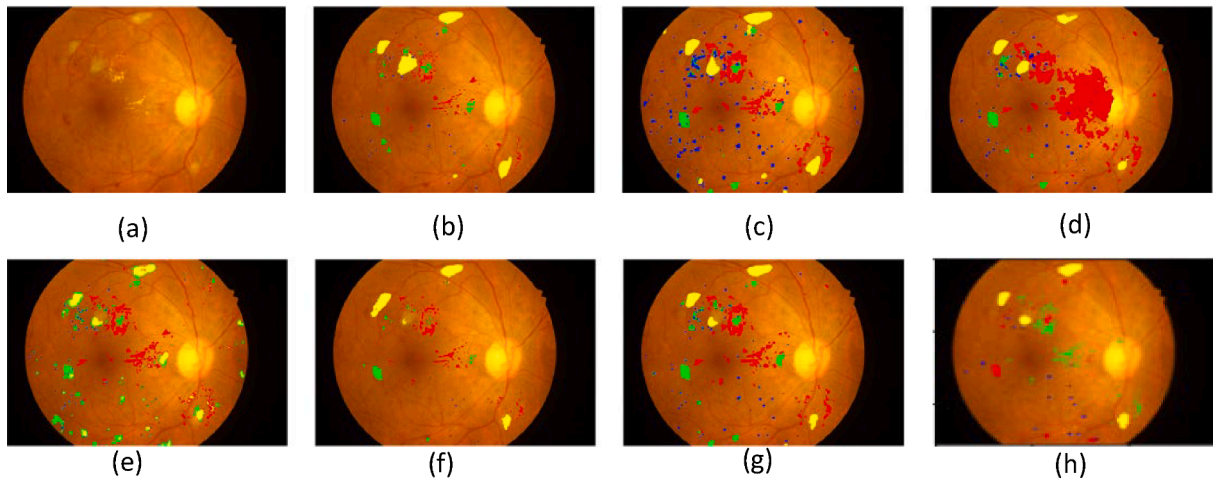


Fig. 15. Comparison of proposed model with previously existing models as per [36] (a) Input Image (b) Ground truth (c) HED (d) FCRN (e) CASENet (f) DeepLab v3+ (g) L-seg (h) RILBP-YNet.

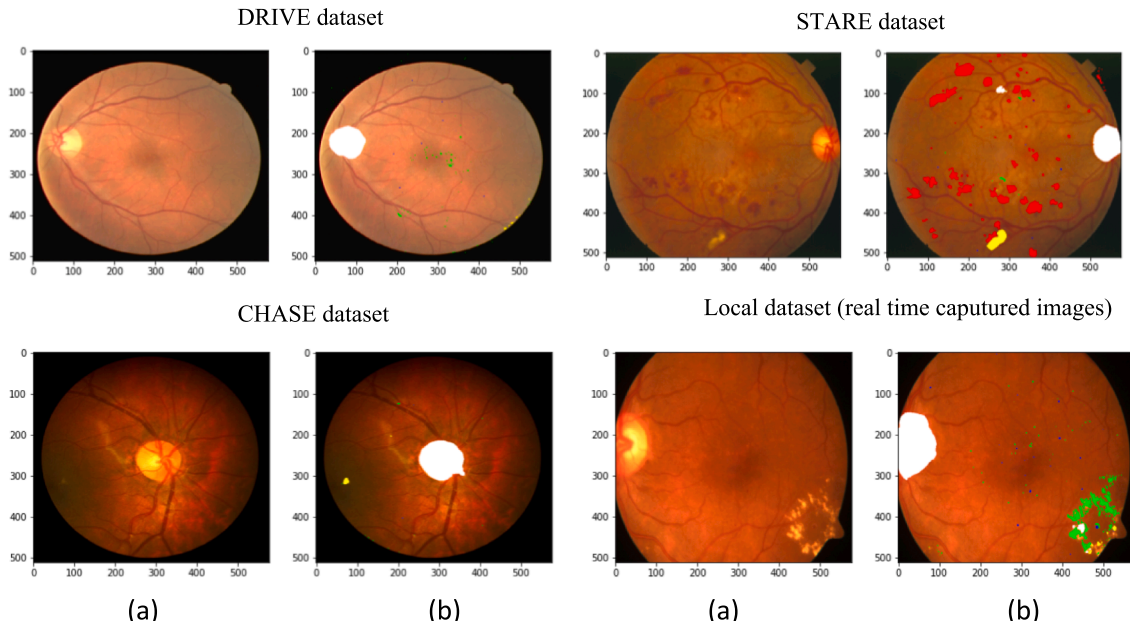


Fig. 16. A random sample images from different datasets like DRIVE, STARE, CHASE, and local dataset (real time captured images) for qualitative analysis of Proposed RILBP-YNNet (a) Input image from each dataset (b) segmented output of (a).

Table 7

Ablation Studies to illustrate the impact of each component of the proposed RILBP-YNNet by evaluating on IDRiD dataset in terms of dice coefficient, parameters size and training time. (* Bold values indicate the highest values.).

Dataset	Models	ME	HE	EX	SE	OD	Parameters	Training time
IDRiD	Residual without LBP	0.320	0.394	0.624	0.479	0.831	1.2 M	2.5 h
IDRiD	Residual with LBP (parallel encoder)	0.343	0.431	0.633	0.711	0.926	1.7 M	2.5 h
IDRiD	Inception without LBP	0.375	0.423	0.680	0.671	0.879	3.7 M	2.7 h
IDRiD	Inception with LBP (parallel encoder)	0.404	0.535	0.741	0.737	0.942	6.6 M	3.6 h
IDRiD	RILBP	0.434	0.525	0.751	0.786	0.966	4.8 M	2.8 h

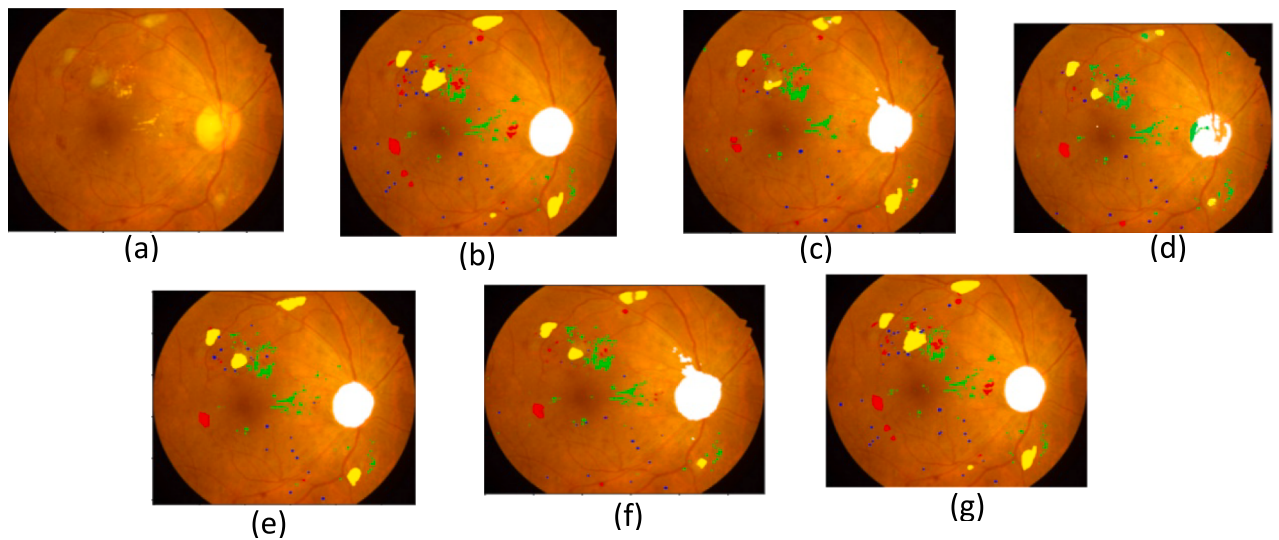


Fig. 17. Ablation studies (a) Input Image (b) Ground Truth (c) Residual with LBP (d) Residual without LBP (e) Inception with LBP (f) Inception without LBP (g) RILBP.

the values of sensitivity, specificity, AUC, and F1 score are increased by at least 50 % for each lesion as shown in Table 9.

The teams that participated in the IDRiD Challenge [9] worked on single lesion segmentation and used the same architecture with different hyper-parameters for segmenting each lesion. But this involves

complexity in loading different networks for different lesions. To circumvent this problem, the proposed network performs multi-class segmentation of lesions simultaneously in one iteration. The performance of our model is compared with the teams that participated in the IDRiD Challenge, and the results are depicted in Table 3. From Table 3, it

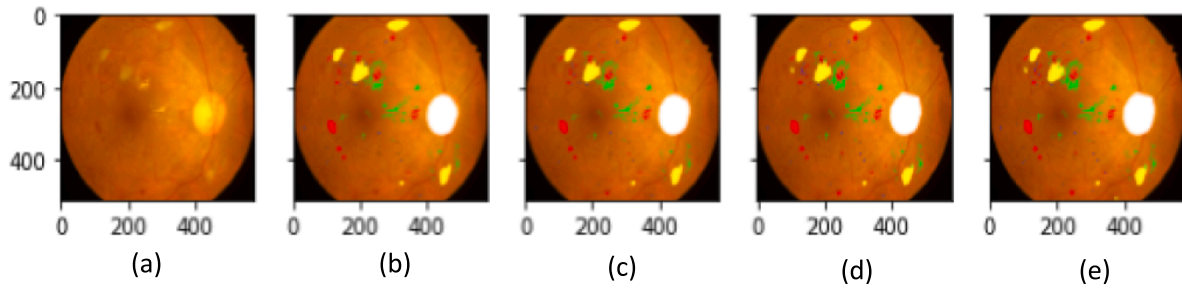


Fig. 18. Ablation Studies to illustrate the impact of loss functions (a) Input Image (b) Ground truth (c) Predicted label with CCE (d) Predicted label with Jaccard loss (e) Predicted label with weighted combination of Jaccard and CCE loss weights.

Table 8

Ablation Studies to illustrate the impact of loss function on proposed RILBP-YNet by using IDRiD dataset in terms of dice coefficient. (* Bold values indicate the highest values.).

Models	ME	HE	EX	SE	OD
CCE	0.281	0.456	0.6840	0.5350	0.9610
Jaccard Loss	0.358	0.5880	0.7350	0.7490	0.9540
JL + CCE	0.434	0.5460	0.7510	0.7860	0.9660

Table 9

Quantitative comparison between serial (S) and parallel (P) structure of convolutional layers in encoder block of RILBP-YNet.

Lesion	Type	Se	Sp	AUC	F1
MA	S	5.14	99.89	16.52	8.82
	P	28.02	99.96	38.04	36.02
HE	S	10.59	99.33	23.93	13.31
	P	57.95	99.69	69.14	66.54
EX	S	41.5	99.71	58.07	53.22
	P	72.15	99.77	78.58	76.32
SE	S	23.01	99.92	36.78	29.94
	P	58.55	99.96	72.67	49.88

is evident that RILBP-YNet shows better competitive results with an average AUC score of 0.6586 vs. 0.6469 of the top performers. Earlier, due to different input sizes and depths of modified UNet used by the teams of IDRiD, the single lesion segmentation was efficient. However, the problem of loading multiple models concurrently during testing is a cumbersome task. The proposed model is compared with previously developed U-Net variants, and the results are shown in Table 4. The proposed model outperformed the baseline models in segmenting all lesions except for MAs due to the small size of MAs. This makes it difficult to maintain the flow of features to higher levels of the network.

To evaluate the efficacy, the proposed network, RILBP-YNet, is trained and tested on different datasets like e-ophtha to validate the efficacy of the network on a diverse set of fundus images. Also, the e-ophtha dataset has only two kinds of lesions present in the fundus images. Further, to estimate the performance of the proposed model on other datasets, it is tested using different datasets such as DRIVE, CHASE, STARE, and locally collected real-time datasets, and respective results are depicted in Fig. 16. These results depicted that the proposed model has significantly detected the lesion pixels along OD pixels with an average dice coefficient of 95 % for OD, 70 % for SE, 75 % for EX, 50 % for HE, and 35 % for ME respectively. The RILBP-YNet can simultaneously segment multi-class lesions, but the capability to segment minute lesions like MAs is less than others. Due to the lack of proper pixel-level annotated datasets and the nature of the lesions, i.e., small sizes, the proposed network's performance is constrained. As a future work, a proper pixel-level annotated dataset will be generated with the help of our collaborating hospital, Sankar Foundation Eye Hospital to further train the network to efficiently segment all lesions from any

fundus image.

The proposed network, RILBP-YNet, performs better segmentation of HEs and SEs compared to the existing lesion segmentation networks. Xue et al. [56] developed two individual Mask-RCNNs to perform the task of MA and EX segmentation and achieved better segmentation results. Also, the Mask-RCNN is trained with different patch sizes of the input image to obtain separate models for segmenting lesions. Similarly, Yan et al. [55] developed Local-Global U-Nets trained using both the input image and patches of the input image to improve segmentation results. The generated individual patched segmentation maps are then combined to obtain a single segmentation mask for each type of lesion. Unlike the existing networks, the proposed network RILBP-YNet considers the entire input image and generates the segmentation maps for individual lesions at a time. This eliminates the need for storing and loading multiple models during testing in real time.

From Table 5, it is evident that the proposed RILBP-YNet outperformed the existing networks in terms of specificity and accuracy in segmenting every lesion. Also, the F1 score of the proposed network is very close to Sarhan et al. [54] in MA segmentation. Similarly, in HE segmentation, the proposed network has achieved first place in Se, Sp, Acc, and F1 and stood at second place in the AUPR score but very close to Yan et al. [55]. The RILBP-YNet produced better results in EX segmentation and outperformed all other networks in SE segmentation. The time complexity of the proposed model is estimated using the date-time libraries and it is observed that the proposed model can segment all lesions in less than 5 s from any testing image of various datasets like IDRiD, e-ophtha, DRIVE, CHASE, STARE, and local dataset.

6. Conclusions

The deep learning network RILBP-YNet has provided a novel path for segmenting simultaneous retinal lesions. The network employed Local Binary Patterns for extracting textural features to enhance the segmentation results. The proposed model not only segments DR lesions but also segments the optic disc more precisely with an mAUC score of 0.7108. The effective segmentation results of the proposed network mimic the experts in examining DR-affected patients. The proposed model outperformed the earlier state-of-the-art segmentation models. To further expand the proposed work, the RILBP-YNet can be trained with different combinations of LBPs and depths of the network. With the increased depth of the network, the receptive field of neurons increases and can extract global semantic features. As a result, it may segment the minute lesions more precisely. Other texture descriptors can be used in place of LBPs to train and improve the performance of the network. Further future work can be aimed to develop an efficient local dataset with pixel-level annotated labels for robust training of the proposed model. In addition to this, the proposed RILBP-Ynet will be implemented in diverse medical imaging fields like brain tumor detection using MRI, lung cancer detection using CT scans, etc as future work.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

CRediT authorship contribution statement

Geetha Pavani P.: Conceptualization, Software, Formal analysis, Writing – original draft. **B. Biswal:** Resources, Investigation, Project administration, Supervision, Data curation, Visualization. **Tapan Kumar Gandhi:** Software, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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